

SHELBYVILLE, IN, USA

WMO: 724356

Lat: 39.578N

Lon: 85.803W

Elev: 245

StdP: 98.42

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53866

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.2	-12.9	-21.0	0.6	-15.1	-18.0	0.8	-12.2	12.0	1.2	11.0	1.2	3.0	300	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.9	23.6	31.8	23.1	30.6	22.5	25.4	30.7	24.6	29.6	23.9	28.5	4.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.2	28.6	23.0	18.2	27.6	22.4	17.6	26.8	79.1	30.7	75.7	29.6	72.6	28.6	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.1	8.2	-20.7	34.4	4.6	2.1	-24.0	35.9	-26.8	37.1	-29.4	38.3	-32.7	39.8		
(5)			-21.2	26.8	4.4	0.8	-24.3	27.4	-26.9	27.9	-29.4	28.3	-32.6	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-1.6	0.0	6.2	12.6	18.3	22.8	23.8	23.0	19.8	13.2	6.6	1.3
	DBStd	10.36	6.64	6.43	6.21	5.19	4.78	3.17	2.96	2.77	3.98	5.20	5.25	5.85
	HDD10.0	1264	363	286	153	32	2	0	0	0	0	26	127	275
	HDD18.3	2849	619	514	379	182	63	4	1	2	29	176	351	528
	CDD10.0	2076	3	5	34	110	259	383	428	404	295	123	26	5
	CDD18.3	619	0	0	2	9	61	138	171	148	74	16	1	0
	CDH23.3	5511	0	1	14	93	534	1177	1583	1274	693	138	3	0
	CDH26.7	1882	0	0	2	10	160	414	592	434	237	33	0	0
Wind	WSAvg	3.8	4.6	4.5	4.5	4.7	3.9	3.3	2.6	2.3	2.7	3.6	4.1	4.3
Precipitation	PrecAvg	1107	77	63	85	109	129	128	108	88	80	75	86	78
	PrecMax	1411	230	131	165	215	224	278	216	152	188	178	189	166
	PrecMin	850	22	22	24	41	36	38	46	27	2	15	29	15
	PrecStd	169	48	33	37	45	52	59	45	34	51	42	39	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	19.4	25.4	28.0	32.0	33.6	35.5	34.0	33.6	30.1	22.7	17.6
		MCWB	14.5	13.9	17.2	17.8	21.8	23.3	25.0	23.9	22.1	20.7	15.8	14.3
	2%	DB	13.7	15.2	21.9	26.2	30.1	32.2	32.9	32.1	31.4	26.9	19.6	14.8
		MCWB	11.8	12.0	15.5	16.9	20.4	22.7	24.4	23.6	22.0	19.0	14.7	12.4
	5%	DB	11.1	12.3	18.3	23.9	28.6	30.9	31.5	30.7	29.4	24.4	17.3	12.1
		MCWB	9.5	9.6	13.6	16.2	20.2	22.2	23.8	23.2	20.9	17.5	13.6	10.2
	10%	DB	7.9	9.1	16.2	21.5	26.6	29.2	30.0	29.1	27.4	22.1	15.1	9.7
		MCWB	5.9	6.3	12.2	14.9	19.3	21.4	23.1	22.5	20.0	16.6	11.5	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.1	18.0	20.3	23.0	25.4	26.8	26.3	24.3	22.2	17.7	15.2
		MCDB	16.5	18.3	24.5	25.9	28.8	30.9	32.3	31.4	29.8	27.8	20.8	16.8
	2%	WB	12.2	12.6	16.2	18.6	22.0	24.5	25.7	25.2	23.5	20.7	15.8	12.7
		MCDB	13.6	14.7	20.4	23.2	28.2	29.7	31.0	30.3	29.0	25.3	18.3	14.3
	5%	WB	9.7	9.9	14.6	17.2	21.2	23.6	24.9	24.3	22.5	19.0	14.0	10.3
		MCDB	11.1	12.3	17.7	21.5	26.8	28.7	29.9	28.7	27.4	22.7	16.9	11.8
	10%	WB	6.0	6.5	12.5	15.9	20.3	22.7	24.0	23.6	21.4	17.3	11.9	7.8
		MCDB	7.9	9.1	15.7	20.5	25.0	27.5	28.6	27.6	25.4	21.2	14.7	9.3
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.7	10.1	11.5	10.7	10.6	10.7	11.2	12.3	11.4	9.8	7.6
		MCDBR	10.4	11.8	13.0	14.1	12.8	12.4	11.8	12.5	14.5	13.8	12.5	10.1
		MCWBR	8.8	9.5	8.1	7.5	5.5	5.2	5.0	5.9	6.2	7.1	8.7	8.7
	5% WB	MCDBR	9.7	11.1	11.4	10.8	10.8	10.2	10.3	10.5	11.5	11.0	10.6	9.4
		MCWBR	9.1	9.7	7.6	7.0	5.3	5.1	5.1	5.6	5.5	6.2	8.3	8.7
Clear-Sky Solar Irradiance	taub	0.306	0.329	0.351	0.406	0.445	0.449	0.460	0.455	0.417	0.360	0.323	0.304	
	taud	2.429	2.337	2.359	2.213	2.152	2.187	2.156	2.157	2.244	2.405	2.457	2.449	
	Ebn at Noon	863	886	904	869	839	834	820	813	822	841	835	837	
	Edh at Noon	82	103	112	138	150	146	149	145	125	95	79	75	
All-Sky Solar Radiation	RadAvg	1.77	2.56	3.63	4.78	5.43	6.15	5.98	5.48	4.51	3.19	2.14	1.47	
	RadStd	0.18	0.30	0.33	0.39	0.47	0.37	0.36	0.30	0.39	0.30	0.24	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+68

Nomenclature: See separate page